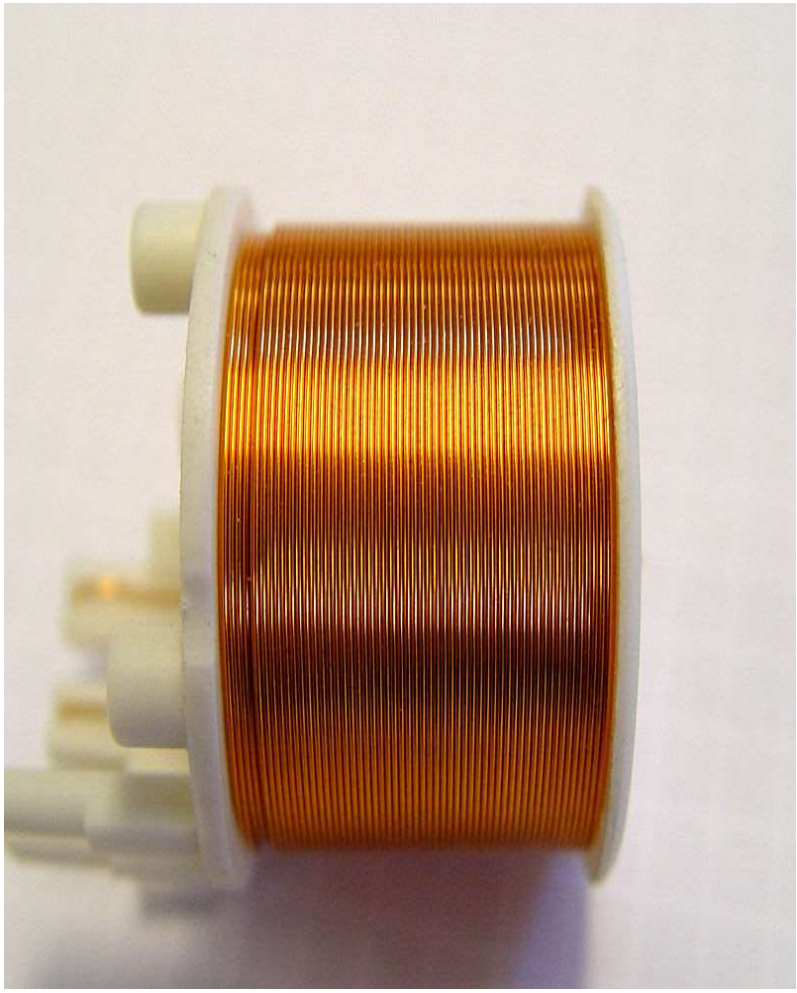
Electrical Workshop Practice II

Chapter One

ELECTRICAL APPARATUS COIL

Contents

- Concepts of apparatus coil and their purpose
- Winding Insulating Materials
- Lab Observation



➢ Lab & Workshop Safety

- Before you can use equipment and machines or attempt practical work in a workshop or a lab, you must understand basic safety rules.
- These rules will help keeping you and others safe in the workshop.
- Following safety rules will also *prevent* you and others from accidents, injuries as well as death.

Insulating Materials (Winding)

- Insulating materials:
  - ➢ Offers very high resistance to the flow of current
  - ➢ Used to keep the current in its proper path along the conductor

Classification of insulating materials

Classification according to substances and materials.

- **Solid Insulating Materials** (Inorganic and organic)
  - Mica, wood, slate, glass, porcelain, rubber, cotton, silk, rayon, paper and cellulose materials etc.
- **Liquid Insulating Materials** (Oils and Varnishes)
  - Refined hydrocarbon minerals oils, Linseed oil, spirit and synthetic varnishes, etc.
- **Gaseous Insulating Materials**
  - Dry air, carbon-dioxide, argon, nitrogène, etc.

• Characteristics of Good Insulating Material

- Very high insulation resistance.
- High dielectric strength.
- Low thermal expansion.
- Non-inflammable when exposed to arcing.
- Resistant to oils or liquids, gas fumes, acids and alkalies.
- Should have no deteriorating effect on the material, in contact with it.
- Good thermal conductivity.
- High mechanical strength
- High thermal strength.
- Should be resistant to thermal and chemical deterioration.
- Should be resistant to moisture absorption.

➢ Insulation Application in Electrical Winding

- Between conductor /coils and earth (phase-to-earth),
- Between conductor /coils of different phases (phase-to-phase),
- Between turns in a coil (inter-turn) and
- Between the coils of the same phase (inter-coil).

➢ Properties of Conductors

- **Electrical properties**
  - Highly conductive
  - Low Resistivity
  - Electrical Energy dissipated as heat must be low
  - Temperature to resistance ratio must be low
- **Mechanical properties**
  - high ductility
  - High tensile strength(withstand stress and strain)
  - Solderability – the joint should have low contact resistance
  - Resistance to corrosion*
  - Easy to fabricate

➢ Coil Winding Basics (Terminologies)

- ***Coils*** are **loops** in a wire, but when electricity flows through the wire interesting things happen.
- *For example*, if there is an iron core in the loops you have an electromagnet (or a solenoid).
- The wire or conductor which constitutes the coil is called the **winding**.
- Each loop of wire is called a **turn**.
- In windings in which the turns touch, the wire must be insulated with a coating of nonconductive insulation such as **plastic** to prevent the current from passing between the wire turns.

Winding Wires

- They are solid wires
- Insulated with varnish*
- Used in transformer, inductor, motor, generator, speaker wires and etc

Wire

- May be cylindrical or rectangular cross-section
- Used to carry electricity and signals (information)
- Sizes are determined by standard gauge values

➢ *Coil Wire Type for Apparatus Winding*

Enameled Round Copper Wire & Enameled Round Aluminum Wire

- A fundamental part of a winding in every electrical machine or apparatus.
- Compared to fibrous insulations, enameled copper & aluminum wire offers a favorable **space saving factor** together with a high level of breakdown voltage.
- The main application of Enameled Copper & Aluminum Wires are in **winding of Transformers** and **Motors**.



- Enameled Aluminum Wire or Enameled Copper Wire is a wire coated with a thin layer of enamel (**varnish**) insulation to prevent the wire surfaces from being in ***a short circuit*** when wound into coils.
- Magnetic flux is created when current flows through the coil.
- They are used in various Electrical Applications due to its superior ***Electrical, Thermal and Mechanical Properties***.
- They are thin in size, super conductive and highly resistive to corrosion.

➢ *Properties of Enameled Wires*

- High dielectric strength
- Very low dissipation factor
- Remaining reasonably constant at high frequencies or under humid conditions
- No mechanical or chemical stripping required.
- Good resistance to abrasion.
- Good flexibility and adhesion to the conductor smaller coefficient of friction,
- Higher mechanical intensity, Excellent heat resistance.
- Good solderability.

The End!